General classification of viruses

↓

Baltimore classification
(David Baltimore — 1970s)

↓

classification based on:

- type of genome present
- strategy of replication

↓

Class (seven)

1. dsDNA
2. ss DNA
3. dsRNA
4. +ve sense ssRNA
5. -ve sense ss RNA
6. RNA viruses that reverse transcribe
7. DNA viruses that reverse transcribe

Double stranded RNA viruses:

- 2 Families
- 8 genuses
- 1 to 9 to 12 segments of dsRNA

Single stranded RNA viruses:

- Families — 7
- Genuses — 15
- can be +ve sense / -ve sense

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Picorna Virus

Size: 22 to 30nm

Shape: Icosahedral

Composition:

- capsid composed of 60 protein subunits
- each protein subunit consist 4 polypeptide (VP1 – VP4)
- no essential lipids

Characteristics:

- single stranded RNA virus
- positive sense
- non-enveloped
- resistant to ether/chloroform
- insensitive to non-ionic detergents
- some unstable at pH 7, many less stable at low ionic strength
- Inactived by light / photodynamic dyes (i.e: neutral red / proflavin)
- Thermal stability varies by divalent cations.

Classification:

↳ 80 species in this family → 35 genera

Human affecting genera:

1. Enterovirus — Gut
2. Rhinovirus — upper respiratory tract
3. Hepatovirus — intestine, liver
4. Parechovirus
5. Kobuvirus

Replication (occur in cytoplasm):

- mRNA encodes RNA dependent RNA polymerase.
- Polymerase synthesize complementary -ss RNA.
- RNA acts template to make more copies.

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Picorna Virus Genome

- 7.2kb in human rhinovirus
- 8.5kb in foot-and-mouth disease virus

→ single stranded +ve sense RNA

5' end of RNA

- long untranslated region of UTR
- "clover leaf" secondary structure called as interal ribosomal entry site (IRES)
- important in translation, virulence and possibly encapsidation
- contain VPg protein attached covalently
- VPg protein provides hydroxyl group to which first oligonucleolide link.

3' end of RNA

- shorter region of UTR
- necessary for -ve strand synthesis during replication
- 3' end covalently modified by polyadenylation

Dieases caused by Picorna Virus

- aseptic meningitis
- encephalitis
- common cold
- foot-and-mouth disease
- conjugtivitis
- hepatitis A
- herpangina
- myositis
- myocarditis

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Togavirus

- Family – Togaviridae
- Two genera: Alpha virus, Rubivirus

Alphavirus → arthropod borne disease
- in humans encephalitis, arthiritis

Rubivirus → common childhood disease
- Rubella
- found globally

Properties:

- single, linear, monopartite +ve dsRNA
- lipoprotein envelope
- genome size 11kb
- virion size 70nm in diameter
- spherical/pleomorphic shape
- glycoprotein projections — spikes
- icosahedral capsid — 40nm in diameter

Togavirus Genome:

- 9700 – 11800 nucleotide long
- capped and polyadenylated (5' cap and 3' tail)

Replication:

① attachment via viral E glycoprotein — endocytosis.
② fusion of virus to host endosomal membrane.
③ release of virus RNA into host cytosol.
④ +ssRNA translated into polyprotein which cleave into non-structural proteins necessary for synthesis of RNA.
⑤ Replication in cytoplasm → dsRNA synthesized

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Flavivirus

Morphology:

- virion 40-60nm in diameter
- spherical shape
- lipid envelope — 2 to 3 glycoproteins
- capsid — single protein

Host Range:

- mammals, arthropod-borne

Taxonomy: 4 genera (89 species)

Genome:

- non-segmented RNA
- +ssRNA
- 9 – 13kb
- 5' cap, 3' polyA-tail

Genome Organization:

- long open reading frame (ORF) at 5' end
- non-coding regions (NCRs) at 3' end
- NCRs form secondary structures required for genome replication & translation
- Flavivirus (except few species) produce unique, small non-coding RNA from 3'-NCRs
  - essential for virus replication in cells
  - modulate pathogenicity in animals
- viral proteins synthesized as part of polyprotein
- polyprotein translationally cleaved by viral and cellular proteases.
- structural proteins present in N-proximal portion of polyprotein.
- non-structural proteins in remaining
- latter includes serine proteases, RNA helicase, and RdRp.

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Flavivirus Genome Replication

↳ occurs in cytoplasm

- occurs by budding through intracellular membrane
- virus particles transported in cytoplasmic vesicles via secretory pathway
- viral genome released by exocytosis
- replication in cytoplasm occurs
- release of viral RNA via exosomes

Examples of Flavivirus

1. Dengue Fever Virus
2. Yellow Fever Virus
3. Hepatitis C Virus
4. West Nile Virus

Two Group Hypothesis

Group 1: non-enveloped, 60-80nm diameter, icosahedral symmetry, several concentric capsid layers, dsDNA genome.

Group 2: enveloped, 50-60nm diameter, +ssRNA genome

Medical Importance

- serologically related to each other
- antibodies of one useful against others
- antigenic variability of Hep-C proteins
- 17D vaccine — most efficient vaccine ever

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CORONA VIRUS

Host Range: Avian / Mammalian Species

Structure:

- enveloped — club shaped proteins spikes
- spherical / pleomorphic
- 120nm in diameter
- capsid composed of matrix protein

Taxonomy:

Order ——— Nidovirus
1 Family ——— Coronaviridae
2 Subfamily ——— coronavirinae / torovirinae

Diseases caused by corona virus:

① common cold
② severe acute respiratory syndrome (SARS)
③ middle east respiratory syndrome (MERS)

Toronavirinae ——→ helical, donut shaped nucleocapsid

Genome:

- +ssRNA genome
- 26-32kb genome
- capped and polyadenylated
- leader RNA at 5' end of genome is also present at the end of subgenomic RNAs.
- leader RNA is 65-89 bp long.

Replication

① attachment viral S protein ——→ endocytosis (virus entry)
② Fusion of virus with endosomal membrane ——→ genome release
③ Synthesis & proteolytic cleavage of replicase polyprotein
④ dsDNA synthesized from +ssRNA
⑤ dsDNA transcribed into new +ssRNA
⑥ synthesis of structural protein by subgenomic mRNA
⑦ assembly and budding
⑧ new virus released by exocytosis

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RETROVIRUS

Structural Components

- outer envelope — lipid bilayer with spikes
- embedded envelop proteins
- capsid — protein shell
- 2 RNA molecules
- reverse transcriptase enzyme

Transmission

blood / semen / syringes / breast milk / vaginal fluids / pre-seminal fluids / rectal fluids / blood transfusions

Examples → HIV that cause AIDS

Charcteristics

- spherical / icosahedral
- 100nm in diameter
- linear, non-segmented genome
- follows reverse transcription

Genome

- +dsRNA (two molecules)
- 7 to 12kb genome
- capped and polyadenylated (3' end)
- 5' cap have 4 regions:
  R, U5, PBS, L
- terminal non-coding regions
- internal region is coding

SALIENT FEATURES

1. only animal viruses
2. can accomodate changes to its configuration
3. lifecycle begins in nucleus of host cell
4. offer gene therapy
5. long term, stable expression of introduced genes.

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Lifecycle of Retrovirus → two phases:

① early phase
② late phase

Early Phase

- bind to target cells
- cross plasma membrane
- reverse transcribe RNA
- uncoat the cores
- find nuclear membrane
- penetrate nucleus

Late Phase

- Transcription
- Transduction
- Assembly
- Maturation
- Budding
- Release

Replication of Retrovirus

① Infection of virus to host cell
② reverse transcription of viral RNA to cDNA
③ DNA integrated into host chromosomes by the enzyme integrase
④ In this integrated DNA state, virus is called as provirus.
⑤ DNA transcribed to RNA by host enzymes
⑥ RNA re-enter cytosol for translation by host ribosomes

→ gag gene translated to capsid proteins
→ pol gene " " reverse transcriptase
→ env gene " " envelop proteins

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Large Viruses DNA Genome:-

① **Adenovirus:-**

- linear dsDNA
- 30-38Kb genome
- 30-40 genes

Genome

→ terminal sequence of each DNA strand is an inverted repeat of 100-140 bp.
→ denatured single strand form "panhandle"
→ panhandle structures important in replication
→ 55-kDa protein known as "terminal protein" is covalently attached to 5' end of each strand.
→ terminal protein acts as primer for initiating the DNA replication
→ cluster of genes express limited no. of shared promoters.

② Herpesviridae

- large family containing more than 100 members
- eight human heipes viruses
- subfamilies based on nucleotide sequence and biological properties.

Subfamilies

① **Alpha-herpesviridae:** 120-180kbp genome
→ latent infection in sensory ganglion
② **Beta-herpesvirinae:** 140-253 kbp genome
→ restricted host range
③ **Gamma-herpesvirinae:** 105-175kbp genome
- Infection of lymphoblastii cell
- human heipes virus IV and VIII

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SMALL DNA GENOMES

① M-13 Phage

- +ssDNA — circular shape
- encode 10 genes
- filamentous capsid

② Bacteriophage

- rigid genome packaging
- 49kbp genome in virus particle
- substrate and DNA packed into virus head
- DNA cleaved at specific site by phage-coded endonucleases.
- This enzyme leaves 12-50bp overhang ~~of~~ on end of each cleaved strands known as cos-site
- hydrogen bond b/w sticky ends — circler molecule

③ T4 Phage

→ 160kbp dsDNA genome
→ replication produces long concentemers of DNA
→ concentemers cleaved by endonucleases
→ some genes repeated at each end of genome

④ Parvovirus

- linear, non-segmented
- -ssDNA of 5kb genome
- ends of genome have palindromic sequences
  - about 115nt → form hairpins
  - essential for genome replication

2 Genes
- rep gene — encode proteins (transcription)
- cap gene — encode coat proteins

⑤ Polyomavirus

- dsDNA — 5kbp → circular
- supercoiled DNA associated with 4 cellular histones
- six genes

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Positive Sense RNA Plant Viruses

1. Bromo-virus
2. Cucumo-virus
3. Alfamo-virus
4. Pomo-virus
5. Tobamo-virus
6. Peclu-virus
7. Furo-virus
8. Hordie-virus
9. Ouramia-virus
10. Ila-virus

Important
2022—2023

Negative Sense RNA Plant Viruses

1. Borna-viridae
2. Banya-viridae
3. Ortho-viridae
4. Para-viridae
5. Rhabdo-viridae

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Bunyavirus

Structure

- spherical
- enveloped
- 90-100nm in diameter

Genome:

- -ssRNA genome
- 3 RNA segments (large, medium, small)
- 2 non-structural proteins (NSm, NSs)

Functions of segments

↳ encode for four structural proteins

① RNA polymerase —— L segment
② Two glycoproteins (Gn and Gc) —— M segment
③ Nucleocapsid —— S segment

Size of segments

large —— 6.3–12
medium —— 3.5–6
small —— 1–2.2

Replication

- occur in cytoplasm
- RNA transcribed into mRNA
- replication
- maturation by budding at Golgi complex
- reassortment of RNA by closely related members

Genera

① Orthobunyavirus
② Hantavirus
③ Nairovirus
④ Phleovirus
⑤ Tospovirus

Families

① Peribunyaviridae
② Hantaviridae
③ Nairoviridae
④ Phenuiviridae

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Arena Viruses

- transmitted from rodents to humans.
- replication same as bunyavirus.

Structure

- round / oval / pleomorphic
- 60-300nm in size
- enveloped
- ss RNA

→ Infection caused by broken skin / respiratory route.
→ 15 animal arenaviruses (8 humans viruses)

Genome

- ambisense genome
- 3' half is -ve sense (untranslated)
- 5' half is +ve sense (conserved)
- two segments of RNA
  - large segment — 7.2 kb
    - encode Z matrix protein and L polymerase
  - small segment — 3.5 kb
    - encode glycoprotein and nucleoprotein
- genes encoded on each segment separated by non-coding intergenic regions (IGR)

Gene Expression

→ RNA polymearse binds to promoter on each encapsidated segments, transcribe mRNA.
→ Transcription is terminated by strong hairpin at each end of gene.
→ mRNA capped by L protein during synthesis

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Orthomyxoviridae

↳ Ortho – proper
myx — mucus

Morphology

- spherical / pleomorphic
- enveloped, segmented
- 80-120nm -ssRNA
- spikes on influenza A/B
- heat labile, pH variant

Genome

→ 8 separate segments (influenza A and B) containing 10 genes
→ nucleoprotein to form nucleocapsid
→ helical symetry
→ lipid bilayer envelop
→ two surface glycoproteins (hemagglutinin / neuraminidase)
→ inactivated by polar solvents

Influenza C → 7 segments of RNA

- Each segmented genome in a separate nucleocapsid called Ribonucleoprotein complex. (RNPs)

- Terminal sequence of RNA bind trimeric RdRp.

Segments

PA, PB1, PB2 — polymerase
HA — hemagglutinin
NA — Neuraminidase
NP — Nucleoprotein
NS — non-structural proteins
M — membrane proteins

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Replication of Orthomyxoviridae

1. entrance through body fluid
2. bind hemagglutinin-glycoprotein and sialic acid receptor
3. entry by endocytosis
4. protein-polymerase complex
5. protein synthesis
6. replication in nucleus
7. 8 segments formed.

Genera → 7

- Influenza A, B, C, D
- Thogotovirus
- Isavirus
- Quaranjavirus

Diseases

Influenza A — humans / birds
Influenza B — human / seals
Influenza C — human / pigs / dogs
Influenza D — cattle / pigs

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Paramyxoviridae

↳ Para — beyond
myxo — mucus

Properties

- lipid bilayer envelop
- nucleocapsid core
- spherical (150-300nm in diameter)
- pleomorphic / filamentous also possible
- spikes — 8 to 12 nm from surface
- host — vertebrates
- sensitive to heat, radiation, chemicals
- Genera 7, species 49
- -ssRNA genome
- non-segmented genome

Genome

→ 15-16kb genome
→ 6-10 tandemly linked genes
lipid envelope — 2 surface glycoproteins
nucleocapsid — nucleocapsid, phosphor, large proteins
b/w envelope & capsid — matrix protein

- genes encoding structural proteins
- accessory genes
  ↳ found as ORFs

Extracistronic regions

3' leader sequence → transcriptional promoter
5' trailer sequence → 50-161 nucleotide long

Intergenomic Regions

- b/w each gene
- 3 nucleotide long

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Rhabdovirus

↳ Greek word Rhabdo — meaning rod

Properties

- enveloped, non-segmented
- bullet shape
- 180nm long, 75nm wide
- -ssRNA
- example: rabies virus

  ↓

  cause acute infection of central nervous syste
- encode 5 proteins
  1. large protein polymerase (L)
  2. glycoprotein (G)
  3. nucleoprotein (N)
  4. phosphoprotein (P)
  5. matrix protein (M)

Genome

→ 11 to 15kb in size
→ protein complex with viral RNA
→ complex aggregate in cytoplasm of host neurons and form Negri-bodies
→ matrix and glycoprotein associated with envelope
→ G protein forms prostrusions covering envelope

Replication

- attachment via G protein
- penetration via fusion/pinocytosis
- uncoat to RNP
- transcription by mRNA
- translation of viral RNA
- new RNA synthesis
- release by budding

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